BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

PHYSICS PART-I — (Re-appear) — (VERSION A) — SECTION-A

Time Allowed: 20 minutes | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number in provided bubble sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Light year is the unit of:

(A) Speed
(B) Light
(C) Distance
(D) Acceleration

Q2. One femto is equal to:

(A) 10^{15}
(B) 10^{-15}
(C) 10^{-18}
(D) 10^{-9}

Q3. Null vector has a _____ direction.

(A) Zero
(B) Forward
(C) Arbitrary
(D) None of these

Q4. A person walk 10 km north and 20 km east, then the resultant displacement of the person is:

(A) 22.36 km
(B) 22.46 km
(C) 25.23 km
(D) 20.36 km

Q5. _____ remain conserved in elastic collision.

(A) Momentum
(B) Kinetic energy
(C) Total energy
(D) All of these

Q6. One horsepower is given by:

(A) 746 W
(B) 746 J
(C) 746 kW
(D) 746 foot-pound

Q7. The object will be weightless when the elevator falls down with _____ acceleration.

(A) Zero
(B) Positive
(C) 9.8 m s^{-2}
(D) -9.8 m s^{-2}

Q8. Artificial satellite moves around:

(A) Moon
(B) Sun
(C) Star
(D) Earth

Q9. With the increase of temperature, viscosity:

(A) Increases
(B) Decreases
(C) Becomes negative
(D) Remain same

Q10. The pressure will be low where the speed of the fluid is:

(A) Constant
(B) Low
(C) High
(D) Zero

Q11. The heating and cooking of food in microwave oven is an example of:

(A) S.H.M
(B) Damped oscillation
(C) Free oscillation
(D) Resonance

Q12. In an isolated system the total energy of vibrating mass spring system is:

(A) Constant
(B) Variable
(C) Low
(D) High

Q13. Longitudinal waves cannot exhibit the property of:

(A) Diffraction
(B) Polarization
(C) Reflection
(D) Interference

Q14. There is no net transfer of energy by particles of medium in _____ waves.

(A) Longitudinal
(B) Transverse
(C) Progressive
(D) Standing

Q15. The principle of Young's double slit experiment is based on the division of :

(A) Amplitude
(B) Frequency
(C) Velocity
(D) Wavelength

Q16. During a sunny day we see the objects in a class room even when all the electric lights are off, due to _____ of light.

(A) Reflection
(B) Refraction
(C) Diffraction
(D) Interference

Q17. In an isothermal process, internal energy of the system:

(A) Decreases
(B) Increases
(C) Become zero
(D) Remains constant

Q18. A real gas can be approximated to an ideal gas at:

(A) Low density
(B) High density
(C) High Pressure
(D) Low Temperature

PHYSICS PART-I — (Re-appear) — SECTION-B

Time Allowed: 02:40 Hrs | **Maximum Marks:** 40

> Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No.2: Attempt any 10 parts of the following. All parts carry equal marks.

i. Show that the famous "Einstein equation" $E = mc^2$ is dimensionally correct.

ii. What is the minimum numbers of unequal vectors to result into a null vector? Explain with the help of the diagram.

iii. Define equilibrium. And explain with the help of an example that the 1st condition of equilibrium is necessary but not sufficient for a body to be in equilibrium.

iv. What is head-on collision? Explain with the help of an example.

v. Is K.E a vector quantity? Explain.

vi. Explain how many minimum number of geo-stationary satellites are required for global coverage of T.V transmission?

vii. Why is a rifle barrel "rifled"?

viii. Why does the pipe of paper squeezes when air is blown through it.

ix. Is there a connection between F and x in mass spring system? Explain.

x. Estimate the frequencies at which a test tube of 15cm long resonates when you blow across its lips.

xi. How would you justify that light waves are transverse in nature?

xii. What is the difference between interference and diffraction?

xiii. Why is the earth not in thermal equilibrium with the sun?

SECTION-C

Maximum Marks: 27

Note: Note: Attempt any (THREE) questions of the following. All questions carry equal marks.

Q03.

a) What is projectile motion? Derive its expression for: i. Maximum height ii. Time of Summit.

b) What is the angle for which maximum height reached and corresponding range are equal?

Q04.

a) What is conservative force and conservative fields? Show that gravitational field is conservative in nature.

b) The engine of a JF-17 Thunder Fighter develops a thrust of 3000N. What horsepower does it at a velocity of 600 m s^{-1}?

Q05.

a) What is Bernoulli's equation? Show that how it is based on the law of conservation of mass?

b) Water flows through a 1 cm diameter pipe with a speed of 1 m/s. what should be the diameter of the nozzle if the water is to emerge at 2.1 m/s?

Q06.

a) State and explain Doppler's effect. What happens to the apparent frequency: i. When a sound source moves towards the stationary listener? ii. When the source and listener both move away from each other?

b) A refrigerator has a coefficient of performance of 8. If the temperature in the freezer is -23°C. What is the temperature at which it rejects heat?

GROUP – II

PHYSICS (PART-I) — (Fresh) — (Version-B) — SECTION –A

Time Allowed: 20 Min | **Maximum Marks:** 18 | **Paper Code:** BN-XI-XXII-II

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The heating and cooking of food evenly by microwave oven is an example of

(A) SHM
(B) Resonance
(C) Damped oscillations
(D) Free oscillation

Q2. The restoring force of SHM is maximum when particle

(A) Displacement is maximum
(B) Half way between them
(C) Crossing mean position
(D) At rest

Q3. Which one of the following factors has no effect on the speed of sound in gas?

(A) Humidity
(B) Pressure
(C) Temperature
(D) Density

Q4. The fundamental wavelength in a closed organ pipe of length "L" is _____

(A) $\lambda = L$
(B) $\lambda = 2L$
(C) $\lambda = 4L$
(D) $\lambda = L/2$

Q5. The principle of Young's double slit experiment is based on the division of _____

(A) Amplitude
(B) Frequency
(C) Velocity
(D) Wavelength

Q6. The property of light which do not change with the nature of medium is

(A) Frequency
(B) Amplitude
(C) Wavelength
(D) Speed

Q7. The process in which the volume of a system remains cont. is called _____

(A) Isobaric
(B) Isothermal
(C) Adiabatic
(D) Isochoric

Q8. When the speed of your car is doubled, by what factor its kinetic energy increase?

(A) [illegible in source scan]
(B) 4
(C) $\sqrt{2}$
(D) 8

Q9. A real gas can be approximated to an ideal gas at:

(A) Low density
(B) High density
(C) High Pressure
(D) Low temperature

Q10. $(2.0\ m \pm 2.0\%)^3$ = _____

(A) $8.0\ m^3 \pm 1.0\%$
(B) $8.0\ m^3 \pm 2.0\%$
(C) $8.0\ m^3 \pm 5.0\%$
(D) $8.0\ m^3 \pm 6.0\%$

Q11. The precision of the measurement 2050m is

(A) 10m
(B) 10Km
(C) 100m
(D) 100Km

Q12. If $\vec{A}.\vec{B} = 1/2\ AB$ the θ is

(A) 0°
(B) 30°
(C) 60°
(D) 45°

Q13. For what angle the equation $|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$ is correct.

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q14. Range of a projectile on a horizontal plane is same for the following pair of angles.

(A) 15° and 18°
(B) 33° and 57°
(C) 20° and 80°
(D) 52° and 62°

Q15. The area under the velocity-time graph is numerically and dimensionally equal to:

(A) Velocity
(B) Displacement
(C) Acceleration
(D) Time

Q16. The unit of gravitational potential is _____

(A) J Kg
(B) J.m
(C) $J\ Kg^{-1}$
(D) $Kg.\ J^{-1}$

Q17. The relation b/w moment of inertia and torque is

(A) $\tau = I/\alpha$
(B) $\tau = \alpha/I$
(C) $\tau = I\alpha$
(D) None

Q18. At sufficiently high speed the flow of viscous fluid becomes

(A) Un-expected
(B) Streamline
(C) Non-viscous
(D) Turbulent

PHYSICS (PART-I) — (Fresh) — SECTION –B

Time Allowed: 2:40Hrs | **Maximum Marks:** 40

Note: Note: Attempt section B & C accordingly.

Q2. (Marks 40) Q.No 2. Attempt any TEN parts of the following. All parts carry equal marks.

i. Why including more digits in answers, does not make it more accurate?

ii. Why do we not keep bananas in the refrigerator?

iii. How can you explain Brewster's law of polarization?

iv. How would you justify that light waves are transverse?

v. Is it possible for astronauts to talk directly to one another even if they remove their helmets?

vi. What is the difference between progressive and stationary waves?

vii. What will be the frequency of simple pendulum if its length is 1m?

viii. How might one be able to locate the position of nodes and anti-nodes in a vibrating string?

ix. Can specific heat of a gas be zero or infinity? Can specific heat be negative?

x. Is K.E a vector quantity?

xi. Find equation for maximum height of projectile.

xii. Why a particle experiencing only one force cannot be in equilibrium?

xiii. For an answer to be complete, the units need to be specified. Why?

SECTION –C

Maximum Marks: 27

Note: Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Define the dot (scalar) product of two vectors? Give example.

(b) To what height can a 400W engine lift 100Kg mass in 3s?

Q4.

a) Show that total energy is conserved in S.H.M. Explain graphically how mechanical energy interchanges in S.H.M?

b) A mass at the end of a spring describes SHM with T = 0.40S. Find out acceleration when the displacement is 0.04m?

Q5.

a) Explain the speed of Sound in a gas and give all factors which affect the speed of sound in air.

b) The speed of sound in air at 0°C is $332ms^{-1}$. What will be the speed of sound at 22°C

Q6.

a) Explain the interference effect produced by thin film.

b) Find the efficiency of a Carnot heat engine working between the steam and ice points?